oxide; quartz; nepheline syenite; zircon such as in the form of zirconium oxide; buddeluyite; and eudialyte. Mixtures of any of the above particles can be used, including different combinations of organic particles, inorganic particles, or both. The silica can be in any suitable form, such as crystalline, amorphous, or precipitated. The alumina can be used in any of its forms, such as alpha, beta, gamma, delta, theta, tabular alumina, and the like and can be fused or calcined, and if calcined, ground or unground.

The particles listed above are widely commercially available. For example, crystalline silica is available from Reade Advanced Materials; amorphous and precipitated silica from PPG Industries, Inc.; ZEEOSPHERES, silica alumina ceramic alloy particles, from 3M Corporation; silica alumina, such as G200, G-400, G-600, from 3M Corporation; alkali alumina silicate, such as W-210, W-410, and W-610, from 3M Corporation; borosilicate glass, sold as SUNSPHERES, from MoSci Corporation; and quartz and nepheline syenite from Unimin, Inc. Other alumina products are available from Micro Abrasives Corporation as WCA3, WCA3S, and WCA3TO, and from Alcoa as T64-20. Zircon, buddeluyite and eudialyte are commercially available from Aran Isles Corporation, and boron nitride is available from Carborundum Inc. as SHP-605 and HPP-325. It will be appreciated that many commercially available products are actually composites or alloys of one or more materials; such particles are equally within the scope of the present invention.

Microparticles can be added in weight percentages of from 0.1 to 20, such as 0.5 to 15, 0.1 to 10, or 0.1 to 0.5, based on the total weight of the composition. Use of micron sized particles contributes to the mar and/or scratch resistance of the coating, but can also affect appearance in the same manner as described above for the silica nanoparticles used in the present additives. Thus, the determination as to the appropriate amount of microparticles will be made by the user based on such things as the application, the weight percent of silica nanoparticles incorporated through the present additives, and the like.

The additives of the present invention can be added at any time during the formulation of the coating. For example, curable powder coating compositions of the present invention can be prepared by first dry blending the film-forming resin, the present additive, and any of the additional additives described above in a blender, such as a Henschel blade blender. The blender is operated for a period of time sufficient to result in a homogenous dry blend of the materials. The blend is then melt blended in an extruder, such as a twin screw co-rotating extruder, operated within a temperature range sufficient to melt but not gel the components. The melt blended curable powder coating composition is typically milled to an average particle size of from, for example, 15 to 80 microns. Other methods known in the art can also be used.

Alternatively, the present powder compositions can be prepared by blending and extruding the ingredients as described above, but without the present additives. The present additives can then be added as a post-additive to the formulation, such as through a second extrusion process, or by simply mixing the particles into the blended composition, such as by shaking them together in a closed container or using a Henschel mixer.

For liquid coatings, the present additives can be added whenever an aminoplast would normally be added to a composition. For example, the present additives can be added to the coating composition under agitation after the film-forming resin is added. The present additives can also be added under agitation after the solvents.

The coating compositions of the invention can be applied to a variety of substrates, for example automotive substrates such as fenders, hoods, doors and bumpers, and industrial substrates such as household appliances, including washer and dryer panels and lids and refrigerator doors and side panels, lighting fixtures, metal office furniture. Such automotive and industrial substrates can be metallic, for example, aluminum and steel substrates, and non-metallic, for example, thermoplastic or thermoset (i.e. "polymeric") substrates. Other objects made from wood and medium density fiberboard can also be coated with the present compositions.

The powder coating compositions are most often applied by spraying, and in the case of a metal substrate, by electrostatic spraying, or by the use of a fluidized bed. The powder coating can be applied in a single sweep or in several passes to provide a film having a thickness after cure of from about 1 to 10 mils (25 to 250 micrometers), usually about 2 to 4 mils (50 to 100 micrometers). Other standard methods for coating application can be employed such as brushing, dipping or flowing.

The liquid compositions of the invention can also be applied by any conventional method such as brushing, dipping, flow coating, roll coating, conventional and electrostatic spraying. Spray techniques are most often used. Typically, film thickness for liquid coatings can range between 0.1 and 5 mils, such as between 0.1 and 1 mil, or about 0.4 mils.

Generally, after application of the coating composition, the coated substrate is baked at a temperature sufficient to cure the coating. Metallic substrates with powder coatings are typically cured at a temperature ranging from 250° F. to 500° F. (121.1° C. to 260.0° C.) for 1 to 60 minutes, or from 300° F. to 400° F. (148.9° C. to 204.4° C.) for 15 to 30 minutes.

Several liquid formulations can be cured at ambient temperature, such as those using a polyisocyanate or polyanhydride curing agent, or they can be cured at elevated temperatures to hasten the cure. An example would be forced air curing in a down draft booth at about 40° C. to 60° C., which is common in the automotive refinish industry. The ambient temperature curable compositions are usually prepared as a two (2) package system in which the curing agent is kept separate from the composition containing the reactive functional group. The packages are combined shortly before application.

The thermally curable liquid compositions such as those using blocked isocyanate, aminoplast, phenoplast, polyepoxide or polyacid curing agent can be prepared as a one-package system. These compositions are cured at elevated temperatures, typically for 1 to 30 minutes at about 250° F. to about 450° F. (121° C. to 232° C.) with temperature primarily dependent upon the type of substrate used. Dwell time (i.e., time that the coated substrate is exposed to elevated temperature for curing) is dependent upon the cure temperatures used as well as wet film thickness of the applied coating composition. For example, coated automotive elastomeric parts require a long dwell time at a lower cure temperature (e.g., 30 minutes 250° F. (121° C.)), while coated aluminum beverage containers require a very short dwell time at a very high cure temperature (e.g., 1 minute 375° F. (191° C.)).

The coating compositions of the invention are particularly useful as primers and as color and/or clear coats in color-clear composite coatings. The compositions of the invention